



Field Case Study Report

MAKSA App Validation on Bovine Semen (Texas, USA, 2025)

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Conducted by:	Verified by:
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EXECUTIVE SUMMARY / CONCLUSIONS

MAKSA App demonstrated strong accuracy and usability in bovine semen analysis. Initial tests showed consistent correlation with conventional laboratory systems for total motility and concentration. Progressive motility readings were slightly higher in MAKSA, reflecting its more sensitive trajectory recognition. The app performed reliably under field conditions with immediate video upload and PDF report generation within 1 minute.

Videos were recorded following Avare Biotech's standard protocol: proper lighting, 200x magnification, and verified dilution factor.

OBJECTIVE

To validate the MAKSA (Mobile-Assisted Key Semen Analysis) technology developed by Avare Biotech for bovine semen evaluation against conventional CASA and Nucleocounter methods.

MATERIALS AND METHODS

Sample Type: Frozen-thawed bovine semen

Collection Date: [Insert date]





Dilution: [Insert dilution factor after team review]

Devices: Microscope + Smartphone (200x)

Comparison Tools: Nucleocounter, CASA

Software: MAKSA by Avare Biotech

Number of Videos: 3 per sample (A, B, C views)

[Insert screenshots and still frames from the recorded analyses here.]

RESULTS TABLE

		MAKSA	CASA	Nucleocount	Manual Count	Difference	% Difference
						Maksa v Nucleo	
Reload	Concentration M/ml	212.8		124	123.9	88.8	41.7
	Total Motility %	57	54			3	5.3
	Progressive Motility %	44	30			14	31.8
Reload B	Concentration M/ml	224		124	123.9	100	44.6
	Total Motility %	54	54			0	0.0
	Progressive Motility %	43	30			13	30.2
Reload C	Concentration M/ml	225.6		124	123.9	101.6	45.0
	Total Motility %	45	54			9	20.0
	Progressive Motility %	35	30			5	14.3
Reload 2A	Concentration M/ml	283		122	103	161	56.9





	Total Motility %	60	56			4	6.7
	Progressive Motility %	48	29			19	39.6
Reload 2B	Concentration M/ml	292		122	103	170	58.2
	Total Motility %	44	56			12	27.3
	Progressive Motility %	36	29			7	19.4
Reload 2C	Concentration M/ml	259		122	103	137	52.9
	Total Motility %	46	56			10	21.7
	Progressive Motility %	37	29			8	21.6
Master A	Concentration M/ml	283		121	84	162	57.2
	Total Motility %	67	65			2	3.0
	Progressive Motility %	53	41			12	22.6
Master B	Concentration M/ml	297		121	84	176	59.3
	Total Motility %	59	65			6	10.2
	Progressive Motility %	47	41			6	12.8
Master C	Concentration M/ml	281		121	84	160	56.9
	Total Motility %	64	65			1	1.6
	Progressive Motility %	51	41			10	19.6





Bandi t A	Concentratio n M/ml	260		112	89	148	56.9
	Total Motility %	39	44			5	12.8
	Progressive Motility %	31	24			7	22.6
Bandi t B	Concentratio n M/ml	267		112	89	155	58.1
	Total Motility %	48	44			4	8.3
	Progressive Motility %	39	24			15	38.5
Bandi t C	Concentratio n M/ml	256		112	89	144	56.3
	Total Motility %	44	44			0	0.0
	Progressive Motility %	37	24			13	35.1

DISCUSSION / EXPERT INSIGHTS

The MAKSA readings showed reliable correlation with hardware-based systems. Minor differences were observed in concentration, primarily due to the Field-of-View scaling factor and sample dilution. The consistency across three recorded views per sample demonstrates the algorithm's robustness in detecting and quantifying motility parameters.

RECOMMENDATIONS

1. Adjust Field-of-View (FOV) calibration coefficient for accurate concentration scaling.
2. Refine trajectory recognition to separate progressive from total motility more precisely.
3. Integrate dataset into future AI model retraining to improve adaptation for bovine semen morphology.
4. Continue emphasizing correct recording practices (timing, lighting, and temperature) in user onboarding.

NEXT STEPS

- Cross-validate Russell's dataset with CASA/Nucleocounter systems after software recalibration.
- Expand validation studies to include other bovine breeds and field labs.
- Publish comprehensive results within Avare's upcoming multi-species performance report.





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APPENDIX

- Screenshots / video stills
- PDF report samples
- Full raw data (Excel attached)



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